

ABBA4 Implicit: Exterior Projection

One symmetric projection around an unprojected triple jump

1 Unprojected fourth-order base

Let $\mathcal{A}_{q,t}$ be one signed endpoint-time ABBA map on two copies, with no intermediate diagonal projection. Define

$$\gamma = \frac{1}{2 - \sqrt[3]{2}}, \quad \delta = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}.$$

The complete base map is

$$\mathcal{C}_{h,t} = \mathcal{A}_{\gamma h, t + (\gamma + \delta)h} \circ \mathcal{A}_{\delta h, t + \gamma h} \circ \mathcal{A}_{\gamma h, t}.$$

Because $2\gamma + \delta = 1$ and $2\gamma^3 + \delta^3 = 0$, the palindromic composition cancels the leading third-order defect of a symmetric second-order base. The middle duration is negative, and the two copies pass continuously through all three factors.

2 One outer symmetric projection

For

$$E = \begin{pmatrix} I \\ I \end{pmatrix}, \quad P = \frac{1}{2} \begin{pmatrix} I & I \end{pmatrix}, \quad G = \begin{pmatrix} I & -I \end{pmatrix}, \quad N = G^T,$$

the reduced formulation is

$$M(\mu) = \mathcal{C}_{h,t}(Ez + N\mu), \tag{1}$$

$$r(\mu) = GM(\mu) + 2\mu = 0, \tag{2}$$

$$z^+ = P(M(\mu) + N\mu). \tag{3}$$

If J_j is the Jacobian of signed factor j , then $J_{\mathcal{C}} = J_3 J_2 J_1$ in execution order and

$$D_{\mu}r = GJ_{\mathcal{C}}N + 2I.$$

Every residual evaluation traverses all three ABBA factors.

3 Configurations

The sole implementation of `ABBA4Implicit` exposes two projection formulations, Newton or Broyden, physical or fully extended state, and optional physical energy tracking. Fully extended execution always enables energy evolution. The selected values are global to the outer step. In the simultaneous formulation the corrected doubled output joins μ in the unknown; elimination gives the reduced residual above. State extension changes what is duplicated, while physical tracking adds only a triangular $\kappa = k/2$ sidecar. Neither choice introduces intermediate projection.

4 Removed per-map implementation

The current default uses one outer projection. The former `after_each_abba_map` implementation, with three distinct roots and intermediate diagonal re-embedding, has been removed. It defined a different numerical map, since projection and factor evolution do not commute. An explicit request for that placement now raises an error. The legacy single-projection constructor is a compatibility factory for the current default and does not define another method.

5 Properties and limitations

The ideal method is symmetric and designed for order four. Geometric claims refer to the converged projection root; finite nonlinear tolerances perturb the map. Negative internal time increments require the potential and dynamics to be evaluable slightly outside the outer interval. Observer records preserve the three ordered factors so the accepted-map tangent can be reconstructed.

6 Implementation correspondence

The public `ABBA4Implicit` configuration is in `src/simulation/methods/abba/order4_implicit.py`; the exterior-projection residual and numerical kernel are in `src/simulation/methods/abba/projection_outer.py`. The older single-projection module provides compatibility imports only. The common ABBA stages and projection helpers live in the neighboring private modules. The canonical architecture document `../simulation/abba4-implicit-simulation-architecture.md` gives the dimensions, diagnostics, and the common prepared integration lifecycle.